

Light-Cone Quantization:

* Start from Nambu formalism

$$S_{NG} = -\frac{1}{2\pi\alpha'} \int d^2\zeta \sqrt{-g^I}$$

g^I : induced metric which is the pull back

$$dg^I = dX^* (M_{\mu\nu})$$

$$g_{\alpha\beta}^I = \partial_\alpha X^\mu \partial_\beta X^\nu M_{\mu\nu}$$

Define: $\partial_0 X^\mu \equiv \dot{X}^\mu$
 $\partial_1 X^\mu \equiv X'^\mu$

$$g_{\alpha\beta}^I = \begin{pmatrix} \partial_0 X^\mu \partial_0 X^\nu M_{\mu\nu} & \partial_0 X^\mu \partial_1 X^\nu M_{\mu\nu} \\ \partial_0 X^\mu \partial_1 X^\nu M_{\mu\nu} & \partial_1 X^\mu \partial_1 X^\nu M_{\mu\nu} \end{pmatrix}$$

$\therefore$

$$g \equiv \det(g_{\alpha\beta}^I) = (\dot{X}^\mu \bar{X}_\mu) (X'^\mu X'_\mu) - (\dot{X}^\mu X'_\mu)^2$$

$$S_{NG} = -\frac{1}{2\pi\alpha'} \int d^2\zeta \sqrt{(\dot{X}^\mu X'_\mu)^2 - (\dot{X}^\mu \bar{X}_\mu) (X'^\mu X'_\mu)}$$

• Canonical Momenta:

$$P_\mu = \frac{\delta S_{NG}}{\delta (\partial_1 X^\mu)}$$

$$\therefore \frac{\delta S_{NG}}{\delta (\partial_1 X^\mu)} = -\frac{1}{4\pi\alpha'} \int d^2\zeta \frac{1}{\sqrt{-g}} \left(2(\dot{X}^\nu X'_\nu) X'_\mu - 2(\dot{X}_\mu) (X'^\nu X'_\nu) \right)$$

$$\therefore P_\mu = \frac{1}{2\pi\alpha'} \frac{(\dot{X}_\mu (X'^\nu X'_\nu) - (\dot{X}^\nu X'_\nu) X'_\mu)}{\sqrt{(\dot{X}^\mu X'_\mu)^2 - (\dot{X}^\mu \bar{X}_\mu) (X'^\mu X'_\mu)}}$$

Claim:

(i) $P^2 + \frac{1}{2\pi\alpha'} (X')^2 = 0$

(ii) $X'^\mu P_\mu = 0$

Proof:

(i) $P_\mu P^\mu = \left(\frac{1}{2\pi\alpha'} \right)^2 \left(\frac{1}{g^I} \right) \textcircled{a}$

$\textcircled{a}: [\dot{X}^\mu (X')^2 - (\dot{X}^\mu X'_\mu) X'^\mu] [\bar{X}_\mu (X')^2 - (\bar{X}^\mu X'_\mu) X'_\mu]$
 $= \frac{\dot{X}^2 (X')^4 - (\dot{X}^\mu X'_\mu)^2 (X')^2 - (\dot{X}^\mu X'_\mu)^2 (X')^2 + (\dot{X}^\mu X'_\mu)^2 (X')^2}{(\dot{X}^\mu X'_\mu)^2 - (\dot{X}^\mu \bar{X}_\mu) (X'^\mu X'_\mu)}$

$(\dot{X}^\mu X'_\mu)^2 - (\dot{X}^\mu \bar{X}_\mu) (X'^\mu X'_\mu) = (\dot{X}^\mu X'_\mu)^2 + (\dot{X}^\mu X'_\mu)^2 \} (X')^2$

$$= \frac{\dot{X} \cdot \dot{X}' - (\dot{X} \cdot \dot{X}')^2}{|\dot{X} - \dot{X}'|^2 - (\dot{X} \cdot \dot{X}')^2}$$

$$= = = (\dot{X}')^2$$

$$\therefore P^2 = P_\mu P^\mu = -\frac{1}{(2\pi\alpha')^2} (\dot{X}')^2$$

$$\text{ii)} \quad (X')^\mu P_\mu = (\dot{X}^\mu) (\dot{X}_\mu (\dot{X}')^2 - (\dot{X} \cdot \dot{X}') \dot{X}'_\mu) = 0 \quad \checkmark$$

* The above two conditions are the Constants of the system

* We can locally bring $g_{\alpha\beta}^\pm$ to the form

$$g_{\alpha\beta}^\pm = e^\pm \eta_{\alpha\beta}$$

$$\eta_{\alpha\beta} = e^{-\phi} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$$

$$\therefore -e^\phi = \partial_\tau X^\mu \partial_\tau X^\nu \eta_{\mu\nu} = |\dot{X}|^2$$

$$e^\phi = |\dot{X}'|^2$$

$$0 = \partial_\tau X^\mu \partial_\sigma X^\mu \eta_{\mu\nu} \Leftrightarrow \dot{X}^\mu \dot{X}'_\mu = 0$$

$$\text{Thus: } P_\mu = \frac{1}{2\pi\alpha'} \frac{(\dot{X}_\mu (\dot{X}')^2 - \dot{X}'_\mu (\dot{X} \cdot \dot{X}'))}{(-|\dot{X}|^2)^{1/2}}$$

$$= \frac{1}{2\pi\alpha'} [\dot{X}_\mu e^\phi] / e^\phi$$

\therefore We are left with

$$P_\mu = \frac{1}{2\pi\alpha'} \dot{X}_\mu \quad (\text{As in the Poly (Sp) action})$$

* Plugging these into our previous constraints.

$$\text{i)} \quad \left(\frac{1}{2\pi\alpha'}\right)^2 \dot{X}^2 + \left(\frac{1}{2\pi\alpha'}\right)^2 (\dot{X}')^2 = 0$$

$$\dot{X}^2 + (\dot{X}')^2 = 0$$

ii)

$$X'^\mu \left(\frac{1}{2\pi\alpha'}\right) \dot{X}_\mu = 0 \Rightarrow X'^\mu \dot{X}_\mu = 0$$

\therefore New constraints: $\dot{X}^2 + \dot{X}'^2 = 0$

$$X'^\mu \dot{X}_\mu$$

Define: $\pi^\mu \equiv \delta x^\mu$

$$\pi^\mu = -\frac{1}{2\alpha'} \dot{x}^\mu$$

* E.O.M for x^μ : $\partial_\mu \left(\frac{\delta \mathcal{L}_{NG}}{\delta (\partial_\mu x^\mu)} \right) = 0$

$$\partial_\tau \left(\frac{\delta \mathcal{L}_{NG}}{\delta (\partial_\tau x^\mu)} \right) + \partial_\sigma \left(\frac{\delta \mathcal{L}}{\delta (\partial_\sigma x^\mu)} \right) = 0$$

$$\partial_\tau P^\mu + \partial_\sigma \pi^\mu = 0 \quad (x^\mu \text{ E.O.M})$$

$$\ddot{x}^\mu - x''^\mu = 0$$

* Recall that we require δS_{NG} to vanish on $\partial \Sigma$ and as such have:

$$\delta S_{NG} = 0 = [E.O.M] + \int_{\tau_1}^{\tau_2} \frac{\delta \mathcal{L}}{\delta (\partial_\sigma x^\mu)} \delta x^\mu \Big|_{\sigma=0}^{\sigma=\pi}$$

(This is where π^μ comes from). Thus, we must impose

$$\pi^\mu(0) = \pi^\mu(\pi) = 0 \quad (\text{Open-string B.C.'s.})$$

$$\text{or,} \quad (x')^\mu(0) = (x')^\mu(\pi) = 0$$

Light Cone coordinates:

* These two constraints do not fix the full diffeomorphism invariance:

→ To see why we introduce the light cone coordinates

on Σ_g :

on M^D

$$\xi^\pm := \tau \pm \sigma$$

$$x^\pm := \frac{1}{\sqrt{2}} (x^0 \pm x^1)$$

$$\tau = \frac{1}{2} (\xi^+ + \xi^-)$$

$$\sigma = \frac{1}{2} (\xi^+ - \xi^-)$$

This implies: $\partial_\tau = \frac{\partial \xi^+}{\partial \tau} \frac{\partial}{\partial \xi^+} + \frac{\partial \xi^-}{\partial \tau} \frac{\partial}{\partial \xi^-}$

$\partial_\tau = \partial_+ + \partial_-$

$\partial_\sigma = \partial_+ - \partial_-$

* Let us now look at the two constraints in these sets of coordinates

$$\dot{x}^2 + \dot{x}^1 = 0$$

$$\begin{aligned} \dot{x}^\mu \partial_{\dot{x}^\mu} &= (\partial_+ + \partial_-) x^\mu (\partial_+ + \partial_-) x_\mu \\ &= (\partial_+ x + \partial_- x) (\partial_+ x + \partial_- x) \\ &= (\partial_+ x)^2 + (\partial_- x)^2 + 2 x^\mu \partial_+ x_\mu \end{aligned}$$

$$\begin{aligned} (x')^2 &= (\partial_+ - \partial_-) x)^2 \\ &= (\partial_+ x)^2 + (\partial_- x)^2 - 2 x^\mu \partial_+ x_\mu \end{aligned}$$

$$\therefore \dot{x}^2 + \dot{x}^1 = 0 \Rightarrow (\partial_+ x)^2 + (\partial_- x)^2 = 0 \quad (i)$$

$$* \dot{x}^\mu x'_\mu = 0 \Rightarrow (\partial_+ + \partial_-) x^\mu (\partial_+ - \partial_-) x_\mu$$

$$(\partial_+ x)^2 = (\partial_- x)^2 \quad (ii)$$

Together this tells us

$$\begin{aligned} \partial_+ x^\mu \partial_+ x_\mu &= 0 \\ \partial_- x^\mu \partial_- x_\mu &= 0 \end{aligned} \quad (\text{light-cone constraints})$$

* Let $f, g: \Sigma_g \rightarrow \mathbb{R}$ i.e. diffeos:

$$\xi^+ \equiv f(\xi^+)$$

$$\xi^- \equiv g(\xi^-)$$

$$\partial_+ = \frac{\partial \xi^+}{\partial \xi^+} \frac{\partial}{\partial \xi^+}$$

Thus, all we get are the change of coord.
Matrix ∂/∂ which does not change the
constraints.

E.O.M: $\partial_\tau \partial_\sigma X_\mu = 0$

• Under conformal transf. x^μ is a scalar: $x^\mu(\frac{\tau}{\sigma}) = x^\mu(\frac{\tau}{\sigma})$

• We had the B.C.'s:

$$x^{\mu'}(\sigma) = 0 \text{ at } \sigma = 0, \pi$$

Note $\sigma = \frac{1}{2}(\xi^+ - \xi^-) \Rightarrow \xi^+ = \xi^-$ (B.C.)

$\sigma = \pi \Rightarrow \xi^+ - \xi^- = 2\pi$

• We require $\partial \xi^+ / \partial \xi^+ = \partial \xi^- / \partial \xi^-$ at $\sigma = \pi$
— This tells us that f and g are related
on the open string:

M^D : Light-cone coordinates: $x^\mu \rightarrow \{x^+, x^-\}$
 $\mu = 0, \dots, D-1$

$$\left\{ \begin{aligned} x^\pm &= \frac{1}{\sqrt{2}}(x^{D-1} \pm x^0) \\ x^i &\equiv x^\mu \end{aligned} \right. \quad i = 1, \dots, D-2$$

$$U^\pm = \frac{1}{\sqrt{2}}(U^{D-1} \pm U^0)$$

$$V_\pm = \frac{1}{\sqrt{2}}(V_{D-1} \pm V_0)$$

* Metric in $\pm$ coordinates:

$$\eta_{+-} = \eta_{-+} = 1, \quad \eta_{ij} = \delta_{ij}$$

$$S^{\mu\nu} \eta_{\mu\nu} = S^+ U^- + U^- S^+ + \sum_{i=1}^{D-2} S^i U^i$$

* Note: Many others define light cone
coordinates on M^D as $x^0 \pm x^{D-1}$
and thus: $\eta_{+-} = -1$

I'm assuming West uses the above to not deal w/ this $(-)$ sign.

- Since x^μ satisfies the wave eq

$$\partial^\mu \partial_\mu x^\mu = 0$$

x^μ has a general solution:

$$x^\mu = l^\mu(\xi^+) + R^\mu(\xi^-)$$

which has Boundary C's: $x'(\sigma)|_{\sigma=0,\pi} = 0$

$$(\partial_+ - \partial_-)(l^\mu + R^\mu) = 0$$

$$\Leftrightarrow \partial_+ l^\mu = \partial_- R^\mu \text{ at } \sigma = 0, \pi$$

These are the same constraints we found for the conf. transl. $\tilde{\xi}^\pm$:

Proof: $\partial_+ x^\mu = \partial_- x^\mu$ and since $x'^\mu(\tilde{\xi}^+, \tilde{\xi}^-) = x^\mu(\tau, \sigma)$

$$\partial_+ x'^\mu = \partial_- x'^\mu$$

$$\text{RHS: } \partial_{\tilde{\xi}^+}(\tilde{\partial}_+ x'^\mu) = \partial_{\tilde{\xi}^-}(\tilde{\partial}_- x'^\mu)$$

But

$$\tilde{\partial}_+ x'^\mu = \tilde{\partial}_- x'^\mu$$

($\leftarrow x'$ here is not $\frac{dx^\mu}{d\sigma}$)

and so we find

$$\partial_+ f^+ = \partial_- g^-$$

* We now write x^μ as follows

$$x^\mu = x_0^\mu + \tau A^\mu + K^\mu(\xi^+, \xi^-)$$

Recall the E.C.M are $\partial^2 x^\mu = 0 \rightarrow \partial_+ \partial_- x^\mu$

$$\begin{cases} \partial_- x^\mu(\xi^+, \xi^-) = g^\mu(\xi^-) \\ \partial_+ x^\mu(\xi^+, \xi^-) = f^\mu(\xi^+) \end{cases} \quad | \quad \tau = \frac{1}{2}$$

Not sure

$$x^\mu(\sigma, \tau) = x^\mu(\xi^+, \xi^-)$$

$$x^\mu(\sigma, \tau) \approx x^\mu(0, 0) + \tau A^\mu + K^\mu(\xi^+, \xi^-)$$

$$\Leftrightarrow x^\mu(\sigma, \tau) = x_0^\mu + \tau A^\mu + K^\mu(\xi^+, \xi^-)$$

Now, under $\xi^\pm \rightarrow \tilde{\xi}^\pm$

$$\tilde{\xi}^+ = \xi^+ + \epsilon^+$$

$$\tilde{x}^\mu \approx x_0^\mu + \tau A^\mu + K^\mu(\tilde{\xi}^+, \tilde{\xi}^-)$$

$$\begin{aligned}
&= X_0^+ + \tilde{\tau} A^+ + K^+(\tilde{\xi}^+, \tilde{\xi}^-) + (\tilde{\xi}^+ - \tilde{\xi}^+) \partial_+ K^+(\tilde{\xi}^+, \tilde{\xi}^-) \\
&= X_0^+ + \tilde{\tau} A^+ + K^+(\tilde{\xi}^+, \tilde{\xi}^-) + (\tilde{\xi}^- - \tilde{\xi}^-) \partial_- K^+ + (1) \partial_+ K^+ \\
&\quad + \mathcal{O}(\tilde{\xi}^+ \tilde{\xi}^-) \\
&= X_0^+ + \tilde{\tau} A^+ + K^+(\tilde{\xi}^+, \tilde{\xi}^-) + f^+ \frac{\partial K^+}{\partial \tilde{\xi}^+} + g^- \frac{\partial K^+}{\partial \tilde{\xi}^-} + \mathcal{O}(1).
\end{aligned}$$

where

$$\tilde{\tau} = \frac{1}{2}(\tilde{\xi}^+ + \tilde{\xi}^-), \quad A^+ \equiv C$$

thus:

$$\begin{aligned}
\tilde{X}^+ &= X_0^+ + \frac{C}{2}(\tilde{\xi}^+ + \tilde{\xi}^-) + (1) \\
\tilde{X}^+ &= X_0^+ + \frac{C}{2}f^+ + \frac{C}{2}g^- + f^+ \frac{\partial K^+}{\partial \tilde{\xi}^+} + g^- \frac{\partial K^+}{\partial \tilde{\xi}^-} + (1) \\
&\quad + K^+
\end{aligned}$$

$$\textcircled{1} - \tilde{\xi}^+ \partial_+ K^+ - \tilde{\xi}^- \partial_- K^+ + \mathcal{O}(\tilde{\xi}^+ \tilde{\xi}^-)$$

But these terms vanish since $\partial_+ \partial_- K^+ = 0$

$$\begin{aligned}
&= -(\tau + \sigma) \partial_+ K^+ - (\tau - \sigma) \partial_- K^+ \\
&= \tau (-\partial_+ K^+ - \partial_- K^+) + \sigma (\partial_- K^+ - \partial_+ K^+)
\end{aligned}$$

Is this C?

Actually, I think it is.... Note if

$$\begin{aligned}
&\partial_- K^+ = \partial_+ K^+, \text{ then} \\
&C \equiv -(\partial_+ K^+ + \partial_- K^+) \\
&= -2(\partial_+ K^+)
\end{aligned}$$

Which gives us the original form of X^+ defined. Cool.

- This seemed rather circular but we've found a constraint on K^+

$$\partial_+ \partial_- K^+ = 0$$

$$\partial_- K^+ = \tilde{A}^+(\tilde{\xi}^-)$$

$$\partial_+ K^+ = \tilde{B}^+(\tilde{\xi}^+)$$

But these two must be equal

$$\tilde{A}(\tilde{\xi}^-) = \tilde{B}(\tilde{\xi}^+)$$

which can only occur if $\tilde{A} = \tilde{B} = \text{const.}$

$$\partial_+ k^+ = n$$

$$\partial_- k^+ = n$$

$$\Rightarrow k^+ = (\xi^+ + \xi^-) n$$

Recall

$$c = -2(\partial_+ k^+) = -2n$$

$$\therefore n = -c/2$$

$$k^+(\xi^+, \xi^-) = -\frac{c}{2}(\xi^+ + \xi^-)$$

* This is where the solution stated by West in the paragraph below (4.1.22) comes from

$$x^+ \sim (l^+ - \frac{1}{2}c\xi^+) + (k^+ - \frac{1}{2}c\xi^-)$$

Going back to x_0 :

$$\tilde{x}^+ = x_0^+ + \frac{c}{2} \xi^+ + \frac{c}{2} \xi^- + l^+ (-\frac{c}{2}) + (-\frac{c}{2}) \xi^- + c\tau$$

+ k^+

$$= x^+$$

• Not sure where the gauge fixing allows us to choose $k^+ = 0$

This would mean $c = 0$, leaving x_0^+ only.

... Anyway...

$$x^+(\sigma, \tau) = x_0^+ + \tau c$$

• Since $P^\mu = \frac{1}{2\pi\alpha'} \dot{X}^\mu$

$$P^+ = \frac{1}{2\pi\alpha'} c$$

The total momentum:

the total momentum is:

$$P^+ \equiv \int_0^\pi d\sigma PM(\sigma) = \frac{c}{2\pi\alpha'} (\pi - a) \quad \left(\begin{array}{l} \text{total} \\ \text{momentum} \end{array} \right)$$

Open string: $a=0$

$$c = 2\alpha' P^+$$

$$\Rightarrow P^+ = \frac{P^+}{\pi}$$

closed string:

$$P^+ = \frac{c}{\alpha'} \Rightarrow c = \alpha' P^+$$

$$P^+ = \frac{P^+}{2\pi}$$

* For interacting open strings we take
 $0 \leq \sigma \leq \pi\alpha' P^+$
 this leaves $c=2$

Open string:

We can rewrite $P^2 + \frac{1}{(2\pi\alpha')^2} (X')^2 = 0$

using $c = 2\alpha' P^+$ and $X^+(\tau, \sigma) = x_0^+ + c\tau$
 $P^+ = P^+/\pi$

$$P^2 = P^- P^+ + P^+ P^- + \sum_i P^i P^i$$

$$= 2 \frac{P^+}{\pi} P^- + \left(\frac{1}{2\pi\alpha'} \right)^2 \dot{X}_i \dot{X}_i$$

$$X'^\mu X'_\mu = 2 \cancel{X^+ X^-} + X^i X^i$$

$$= 0 + (X^i)' ^2$$

$$\therefore 2 \frac{P^+}{\pi} P^- + \sum_{i=1}^{D-2} \left(P_i^2 + \frac{(X_i')^2}{(2\pi\alpha')^2} \right) = 0$$

$$P^- = - \frac{\pi}{2P^+} \sum_{i=1}^{D-2} \left(P_i^2 + \frac{(X_i')^2}{(2\pi\alpha')^2} \right)$$

Constraints

• $X'^\mu P_\mu = 0$

$$X'^+ P^- + X'^- P^+ + \sum_{i=1}^{D-2} X'^i P_i = 0$$

$$\frac{P^+(X^-)'}{\pi} + \sum_{i=1}^{D-2} X'^i P_i = 0$$

$$(X^-)' = -\frac{\pi}{P^+} \sum_{i=1}^{D-2} X_i' P_i$$

$$X^-(\sigma, \tau) = A(\tau) - \frac{\pi}{P^+} \int_0^\sigma d\sigma' \sum_{i=1}^{D-2} \frac{d}{d\sigma'} X_i'(\sigma', \tau) P_i(\sigma', \tau)$$

• Define the C.O.M coordinate

$$q^-(\tau) \equiv \frac{1}{\pi} \int_0^\pi d\sigma' X^-(\tau, \sigma')$$

$$A(\tau) \equiv (q^- + \tilde{A})(\tau)$$

$$X^-(\sigma, \tau) = q^-(\tau) + \tilde{A}(\tau) - \frac{\pi}{P^+} \int_0^\sigma d\sigma' \left(\right)$$

Integrate over x^- to find $\tilde{A}$:

$$q^-(\tau) = q^-(\tau) + \tilde{A}(\tau) - \frac{1}{P^+} \int_0^\pi d\sigma'' \int_0^{\sigma''} d\sigma' \sum_{i=1}^{D-2} X_i'(\tau, \sigma') P_i'(\sigma', \tau)$$

$$\therefore \tilde{A}(\tau) = \frac{1}{P^+} \int_0^\pi d\sigma'' \int_0^{\sigma''} d\sigma' \sum_{i=1}^{D-2} X_i'(\tau, \sigma') P_i'(\sigma', \tau)$$

* West (4.1.3) [Also Fubini's theorem]

Note: I wasn't able to obtain the following equations via integration by parts, Instead, I use the fundamental theorem of calculus:

Let $G(x) \equiv \int_0^x f(t) dt$, then $G'(x) = f(x)$.

Now,

$$F(x) \equiv \int_0^x du f(u) (x-u)$$

$$f(x) \cancel{(x-x)} + \int_0^x du f(u)$$

$$\int_0^{x'} F(x) dx = F(x')$$

$$= \int_0^{x'} dx \int_0^x du f(u)$$

Note the constant of integration is 0: we

see any calc Book.

conclude:

$$F[x] = \int_0^x dt \int_0^t dz f(z) = \int_0^x dz f(z) (x-z)$$

Thus

$$\int_0^x dt \int_0^t dz f(z) = \int_0^x dz f(z) (x-z)$$

Use this to obtain what West has!!!

$$\begin{aligned} \tilde{A}[\tau] &= \frac{1}{p_+} \int_0^\pi d\sigma \int_0^\sigma d\sigma' \sum_{i=1}^{D-2} \dot{X}_i' P_i \\ &= -\frac{1}{p_+} \int_0^\pi d\sigma' (\sigma' - \pi) \sum_{i=1}^{D-2} \dot{X}_i' P_i \end{aligned}$$

$$\begin{aligned} \therefore X^-(\tau, \sigma) &= q^-(\tau) - \frac{\pi}{p_+} \int_0^\sigma d\sigma' \sum_{i=1}^{D-2} \dot{X}_i'(\tau, \sigma') P_i^i(\tau, \sigma') \\ &\quad - \frac{1}{p_+} \int_0^\pi d\sigma' (\sigma' - \pi) \sum_{i=1}^{D-2} \dot{X}_i'(\tau, \sigma') P_i^i(\tau, \sigma') \\ &= q^-(\tau) + \frac{\pi}{p_+} \int_0^\sigma (1) + \frac{\pi}{p_+} \int_0^\pi (1) \\ &\quad - \frac{1}{p_+} \int_0^\pi d\sigma' \sigma' (1) \end{aligned}$$

→ Swapped limits of integration: $(\int_0^\sigma + \int_0^\pi) = \int_0^\pi$

$$\begin{aligned} \therefore X^-(\tau, \sigma) &= q^-(\tau) + \frac{\pi}{p_+} \int_0^\pi d\sigma' \sum_{i=1}^{D-2} \dot{X}_i'(\tau, \sigma) P_i^i(\tau, \sigma) \\ &\quad - \frac{1}{p_+} \int_0^\pi d\sigma' \sigma' \sum_{i=1}^{D-2} \dot{X}_i'(\tau, \sigma') P_i^i(\tau, \sigma') \end{aligned}$$

(open string)

Closed string:

$$\begin{aligned} X^-(\tau, \sigma) &= q^-(\tau) - \frac{\pi}{p_+} \int_{-\pi}^\sigma d\sigma' \sum_{i=1}^{D-2} \dot{X}_i'(\tau, \sigma') P_i^i(\tau, \sigma') \\ &\quad - \frac{1}{2p_+} \int_{-\pi}^\pi d\sigma' (\sigma' - \pi) \sum_{i=1}^{D-2} \dot{X}_i'(\tau, \sigma') P_i^i(\tau, \sigma') \end{aligned}$$

$$q^-(\tau) = \frac{1}{2\pi} \int_{-\pi}^\pi d\sigma' X^-(\tau, \sigma')$$

Summary: started w/ X^μ, P^μ and used residual Conf.

Symm → choose X^+ to be of the form $c\tau + x_0^+$

$$\bullet \underline{P}^+ = \underline{P}^+(\sigma)$$

then used these to solve for $P^-(\tau, \sigma), (X^-)'(\tau, \sigma)$

• Left w/

$$(x^i, p^i, q^-, p^+)$$

Hamiltonian:

* Generator of time translations:

$$X^+(\tau, \sigma) = x_0^+ + c\tau$$

These translations in τ are directly related to variations in x^+ : $c = 2\alpha' p^+$

$$\delta X^+ = c \delta \tau$$

$$\therefore \delta \tau = \left(\frac{1}{2\alpha' p^+}\right) \delta x^+$$

$$H = \dot{X}^\mu P_\mu - \mathcal{L}$$

Use $\mathcal{L}_{phys}$ to obtain the result:

$$H = -2\alpha' p^+ p^- = -2\alpha' p^+ \int_0^\pi d\sigma P^-(\sigma)$$

Plugging in our value for P^- we find

$$H = -2\alpha' p^+ \int_0^\pi d\sigma \left\{ -\frac{\pi}{2p^+} \sum_{i=1}^{D-2} \left(p_i^2 + \frac{1}{(2\pi\alpha')^2} x_i'^2 \right) \right\}$$

$$H = \pi\alpha' \int_0^\pi d\sigma \left\{ \sum_{i=1}^{D-2} \left(p_i^2 + (x_i' / 2\pi\alpha')^2 \right) \right\}$$

* We now have our given variables and no constraints to solve:

$$\text{Poisson: } \begin{cases} \{x^i(\sigma), p^i(\sigma')\} = \delta^{ij} \delta(\sigma - \sigma') \\ \{q^-, p^+\} = 1 \end{cases}$$

$$\{X^\mu(\sigma), P^\nu(\sigma')\} = \eta^{\mu\nu} \delta(\sigma - \sigma')$$

For $\mu \neq i$:

$$\{X^-(\sigma), P^+(\sigma')\} = \delta(\sigma - \sigma')$$

$$\frac{1}{\pi} \int_0^\pi \{X^-(\sigma), \frac{P^+}{\pi}\} = \frac{1}{\pi} \int_0^\pi d\sigma \delta(\sigma - \sigma')$$

$$\therefore \{q^-, p^+/\pi\} = \frac{1}{\pi} \mathbb{1} \Leftrightarrow \{\pi q^-, p^+\} = 1$$

π : Not sure where this factor has come from....

* Proof of H generating time translations:

i.i) Want $\dot{X}^i = \{X^i, H\}$

$$\begin{aligned} \text{RHS: } \{X^i(\sigma), H\} &= \left\{X^i, \pi\alpha' \int_0^\pi d\sigma' \sum_{j=1}^{D-2} \left(P_j^2 + \left(\frac{\dot{X}_j}{2\pi\alpha'} \right)^2 \right) \right\} \\ &= \pi\alpha' \left\{X^i, \int_0^\pi d\sigma' \sum_{j=1}^{D-2} P_j^2(\sigma') \right\} \\ &= \pi\alpha' \int_0^\pi d\sigma' \sum_{j=1}^{D-2} \{X^i(\sigma), P_j^2(\sigma')\} \end{aligned}$$

$$\begin{aligned} \text{Recall } \{X^i(\sigma), P_j^2(\sigma')\} &= \sum_l \left(\frac{\partial X^i}{\partial x^l} \frac{\partial P_j^2}{\partial p^l} - \frac{\partial P_j^2}{\partial x^l} \frac{\partial X^i}{\partial p^l} \right) \\ &= \sum_l \delta_{il} 2 P_j \delta_{jl} \delta(\sigma - \sigma') \end{aligned}$$

$$\begin{aligned} \therefore \{X^i(\sigma), H\} &= \pi\alpha' \int_0^\pi d\sigma' \sum_{j=1}^{D-2} \sum_{l=1}^{D-2} \delta_{il} 2 P_j \delta_{jl} \delta(\sigma - \sigma') d\sigma' \\ &= 2\pi\alpha' \sum_{j=1}^{D-2} \delta_{ij} P_j \\ &= 2\pi\alpha' P_i \end{aligned}$$

$$\{X^i(\sigma), H\} = 2\pi\alpha' P_i$$

$$\text{But } P_i = \frac{1}{2\pi\alpha'} \dot{X}^i$$

• Thus: $\boxed{\dot{X}^i = \{X^i(\sigma), H\}}$

i.ii) Similarly: $\dot{P}^i(\sigma) = \{P^i(\sigma), H\}$

RHS:

$$\begin{aligned} \{P^i(\sigma), \pi\alpha' \int_0^\pi d\sigma' \sum_{j=1}^{D-2} \left(\frac{\dot{X}_j}{2\pi\alpha'} \right)^2 \} \\ = \pi\alpha' \int_0^\pi d\sigma' \sum_{j=1}^{D-2} \left\{ P^i(\sigma), \frac{1}{\alpha^2} \left(\frac{d}{d\sigma'} X_j \right)^2 \right\} \end{aligned}$$

$$\{P^i(\sigma), \left(\frac{d}{d\sigma'} X_j \right)^2\} = \sum_l \frac{\partial P^i}{\partial x^l} \frac{\partial}{\partial p^l} - \frac{\left(\frac{d}{d\sigma'} X_j \right)^2}{\partial x^l} \frac{\partial P^i}{\partial p^l}$$

$$= \sum_l -2 \delta_{jl} \frac{d}{d\sigma'} \delta(\sigma - \sigma') \delta_{il} \left(\frac{d}{d\sigma'} X_j \right)$$

$$\pi\alpha' \int_0^\pi d\sigma' \left(\frac{1}{\alpha^2} \right) \sum_{j=1}^{D-2} \sum_l \left(-2 \delta_{jl} \frac{d}{d\sigma'} \delta(\sigma - \sigma') \left(\frac{d}{d\sigma'} X_j \right) \delta_{il} \right)$$

$$= -2\pi\alpha' \int_0^\pi d\sigma' \frac{1}{\alpha^2} \frac{d}{d\sigma'} \delta(\sigma - \sigma') \frac{d}{d\sigma'} X^i$$

Int. by parts

$$= 2\pi\alpha' \int_0^\pi \frac{d\sigma'}{(2\pi\alpha')^2} \delta(\sigma - \sigma') \frac{d^2}{d\sigma'^2} X^i$$

$$= (X^i)'' / 2\pi\alpha'$$

$$\therefore \boxed{\dot{P}^i(\sigma) = \{P^i(\sigma), H\} = \frac{(X^i)''}{2\pi\alpha'}}$$

• We can relate H to q^- via:

$$\dot{q}^-|\tau\rangle = \frac{1}{\pi} \int_0^\pi d\sigma' \dot{X}^-(\tau, \sigma')$$

But

$$\begin{aligned} \dot{q}^-|\tau\rangle &= \{q^-(\tau), H\} = \{q^-, -2\alpha' p^- p^+\} \\ &= -2\alpha' p^- \{q^-, p^+\} \\ &= -2\alpha' p^- = -H/p^+ \end{aligned}$$

$\therefore$

$$\boxed{\dot{q}^-|\tau\rangle = -2\alpha' p^- = -H/p^+}$$

- H & p^+ are constants of motion ($\dot{H} = \dot{p}^+ = 0$)
- Hence: $\ddot{q}^- = 0$

$$\boxed{q^-(\tau) = q^-(0) - \frac{H}{p^+} \tau}$$

* Open string: $X^i(\tau, \sigma) = x_0^i(\tau) + \frac{1}{2} \sum_{n=-\infty}^{\infty} x_n^i(\tau) e^{in\sigma}$

$$\rightarrow x_n^i(\tau) = x_{-n}^i = (x_n^i)^*$$

$$P^i(\tau, \sigma) = \frac{1}{\pi} \sum_{n=-\infty}^{\infty} p_n^i(\tau) e^{in\sigma}$$

$$p_n^i = p_{-n}^i = (p_n^i)^*$$

• We previously had:

$$x_0^i(\tau) = x_0^i(0) + 2\alpha' p_i \tau$$

• Substituting P^i and X^i back into the expression for H we find:

$$H = \pi \alpha' \int_0^\pi d\sigma' \sum_{j=1}^{D-2} \left[(\dot{P}_j)^2 + \frac{1}{(2\pi\alpha')^2} (X_j')^2 \right] \quad -K\ell$$

$$= \pi \alpha' \int_0^\pi d\sigma' \sum_{j=1}^{D-2} \left\{ \sum_K \frac{1}{\pi^2} P_K^i e^{+iK\sigma'} \sum_L P_L^i e^{iL\sigma'} + \frac{1}{(2\pi\alpha')^2} \sum_K \sum_L X_K^i e^{iK\sigma} e^{iL\sigma} \right\}$$

$$= \pi \alpha' \sum_{iK\ell} \int_0^\pi d\sigma' e^{i\sigma'(K+\ell)} () + e^{-i\sigma'(K+\ell)} ()$$

Integral survives only if $K = -\ell$:

$$\therefore H = \pi \alpha' \sum_{n=-\infty}^{\infty} \sum_{i=1}^{D-2} \left(\frac{(P_n^i)^2}{\pi^2} + \frac{1}{4\pi^2 \alpha'^2} n^2 (X_n^i)^2 \right) \pi$$

Thus,

$$H = \alpha' \sum_{n=-\infty}^{\infty} \sum_{i=1}^{D-2} \left((P_n^i)^2 + \frac{n^2}{4\pi^2 \alpha'^2} (X_n^i)^2 \right)$$

Ham. in terms of
normal
modes

* The Hamiltonian is nothing more than an ∞ -# of Harmonic oscillators...

→ For $n=0$: we have a zero freq. which corresponds to a free point particle

(Recall for a p.p. $H = \frac{1}{2} (p^\mu p_\mu + m^2)$)

$$\alpha' \sum_{i=1}^{D-2} \left((P_0^i)^2 + \frac{1}{4\alpha'^2} (X_0^i)^2 \right) + \sum_{n \neq 0} ()$$

where $m^2 = X_0^2 / 4\alpha'^2$ and $\phi = \alpha'$

* Thus, Quantization is simply Quant. of Harmonic oscillators: $\sigma \rightarrow \sigma \in (-\pi, \pi)$ $\nearrow$ and extend the range of σ via the boundary conditions

$$X^i(\sigma) = X^i(-\sigma)$$

$$P^i(\tau, \sigma) \equiv \frac{1}{\sqrt{2\alpha'}} (\dot{X}^i(\tau, \sigma) + X'^i(\tau, \sigma)) \quad (\text{Oscillators})$$

$$= \pi \sqrt{2\alpha'} P^i(\tau, \sigma) + X'^i(\tau, \sigma)$$

$$= \sum_{n=-\infty}^{\infty} \alpha_n^i e^{-in\sigma}$$

which has Poisson Brackets:

$$\{x_n^i, \alpha_m^j\} = -i n \delta^{ij} \delta_{n+m,0} \quad | \quad \{x_n^i, p_m^j\} = \delta^{ij} \delta_{n+m,0}$$

$$P^i | \tau, \sigma | = \pi \sqrt{2\alpha'} P^i + \frac{x^{i'}}{\sqrt{2\alpha'}}$$

$$\begin{aligned} \sum_{i=1}^{D-2} P^i P_i &= \left(\pi^2 2\alpha' P^i P_i + \frac{1}{2\alpha'} (x_i')^2 + 2 P^i x_i' \right) \\ &= \sum_{i=1}^{D-2} \left(2\alpha' \pi^2 \left(P^i P_i + \frac{1}{4\alpha'^2 \pi^2} (x_i')^2 + \frac{2}{\alpha' \pi^2} P^i x_i' \right) \right) \end{aligned}$$

"H"

Int.:

$$\int_0^\pi \sum_{i=1}^{D-2} P^i P_i d\sigma = 2\pi H + \int_0^\pi d\sigma 2 P^i x_i'$$

$$\begin{aligned} \int_0^\pi \sum_i P_i x_i' d\sigma &= \int_0^\pi 2 \sum_\ell P_\ell^i e^{i\ell\sigma} \sum_k (ik) x_k^i e^{ik\sigma} \\ &= \sum_\ell 2\pi \sum_\ell P_\ell^i i\ell x_\ell^i \\ &= 0 \text{ as only } \ell=0 \text{ contributes... (I think)...} \end{aligned}$$

$$\int_0^\pi = \frac{1}{2} \int_{-\pi}^\pi$$

Thus:

$$H = \frac{1}{4\pi} \int_{-\pi}^\pi d\sigma \sum_{i=1}^{D-2} P^i P_i$$

In terms of normal modes:

$$H = \frac{1}{4\pi} \int_{-\pi}^\pi d\sigma \sum_{i=1}^{D-2} \left(\sum_n \alpha_n^i e^{in\sigma} \right) \left(\sum_m \alpha_m^i e^{im\sigma} \right)$$

as always $n = -m$

$$H = \frac{1}{4\pi} \left(2\pi \sum_{i=1}^{D-2} \sum_{n=-\infty}^\infty \alpha_n^i \alpha_{-n}^i \right) \text{ or}$$

$$H = \frac{1}{2} \sum_{i=1}^{D-2} \sum_{n=-\infty}^\infty \alpha_n^i \alpha_{-n}^i \quad \text{where}$$

$$\alpha_0^i = \sqrt{2\alpha'} p^i$$

• Note:

$$\alpha' P^\mu P_\mu = 2\alpha' P^+ P^- + \alpha' \sum_{i=1}^{D-2} P_i P_i$$

$$\alpha' P^\mu P_\mu = 2\alpha' P^+ P^- + \frac{1}{2} \sum_{i=1}^{D-2} \alpha_0^i \alpha_0^i$$

$$P_i P_i = \frac{1}{2\alpha'} \alpha_0^i \alpha_0^i$$

West has

$$\begin{aligned}
&= -H + \frac{1}{2} \sum_{i=1}^{D-2} \alpha_0^i \alpha_0^i \\
&= \frac{1}{2} \sum_{i=1}^{D-2} \left(\alpha_0^i \alpha_0^i - \sum_{n=-\infty}^{\infty} \alpha_n^i \alpha_{-n}^i \right) \\
&= \frac{1}{2} \sum_{i=1}^{D-2} \left(\alpha_0^i \alpha_0^i - \alpha_0^i \alpha_0^i - 2 \sum_{n=1}^{\infty} \alpha_n^i \alpha_{-n}^i \right)
\end{aligned}$$

Thus, we find:

$$\alpha^\mu p_\mu p_\mu = - \sum_{i=1}^{D-2} \sum_{n=1}^{\infty} \alpha_n^i \alpha_{-n}^i$$

an extra factor of 2 here
This is a typo.

* Thus, we have mass terms

$$M^2 = - \frac{2}{\alpha'} \sum_{i=1}^{D-2} \sum_{n=1}^{\infty} \alpha_n^i \alpha_{-n}^i$$

The Quantum string in light-cone:

* Apply the Dirac Quant:

$$2\alpha' p_i p_i = \alpha_0^i \alpha_0^i$$

$$\begin{aligned}
[x^\mu(\sigma), p^\nu(\sigma')] &= i\hbar \delta(\sigma-\sigma') \delta^{\mu\nu} \\
[q^-, p^+] &= i\hbar \\
[\alpha_m^i, \alpha_n^j] &= n \delta^{ij} \delta_{m+n,0}
\end{aligned}$$

• Most general ψ as an expansion in α_n^i which are eigen fns of H

$$x^\mu \equiv (x_0^\mu, q^-, z)$$

$$\psi = (\psi(x^\mu) + i\alpha_1^{i\dagger} A_i(x^\mu) + \alpha_1^{i\dagger} \alpha_1^{j\dagger} B_{ij}(x^\mu) + i\alpha_2^{i\dagger} C_i(x^\mu) + \dots) \psi_0$$

where

$$\alpha_n^i \psi_0 = 0 = \langle 0,0 | \alpha_n^i | \psi \rangle, \quad n \geq 1$$

• For the Hamiltonian we find we must add a constant α_0 to account for normal ordering:

$$\begin{aligned}
H &= \frac{1}{2} \sum_{i=1}^{D-2} \alpha_0^i \alpha_0^i + \sum_{i=1}^{D-2} \sum_{n=1}^{\infty} \alpha_{-n}^i \alpha_n^i - \alpha_0 \quad | \quad (\alpha_n^i)^\dagger = \alpha_{-n}^i \\
H &= \alpha' \sum_{i=1}^{D-2} p_i p_i + () - \alpha_0
\end{aligned}$$

Thus:

$$H = \alpha' \sum_{i=1}^{D-2} p_i p_i + \sum_{i=1}^{D-2} \sum_{n=2}^{\infty} (\alpha_n^i)^{\dagger} \alpha_n^i - \alpha_0$$

* Classically, $\alpha' p_{\mu} p^{\mu} = -\frac{1}{\alpha'} \sum \sum \alpha_n^i \alpha_{-n}^i$
will become a differential operator in the
Schrödinger rep:

$$p^{\mu} = -i \frac{\partial}{\partial x^{\mu}}:$$

$$(-\alpha' \partial^2 + H = \alpha' \sum_{i=1}^{D-2} p_i p_i) \psi = 0$$

$$(-\alpha' \partial^{\mu} \partial_{\mu} + \alpha' \sum p_i p_i - \alpha' \sum p_i p_i + \sum_{i=1}^{D-2} \sum_{n=2}^{\infty} (\alpha_n^i)^{\dagger} \alpha_n^i - \alpha_0) \psi$$

$$(-\alpha' \partial^{\mu} \partial_{\mu} + \sum_{i=1}^{D-2} \sum_{n=1}^{\infty} (\alpha_n^i)^{\dagger} \alpha_n^i - \alpha_0) \psi = 0$$

Claim: This is simply the Shrö. eq.

$$i \frac{\partial}{\partial \tau} \psi = H \psi$$

where $i \frac{\partial}{\partial \tau} = (\alpha') (\partial^2 + \sum_{i=1}^{D-2} p_i p_i)$
 $= (\alpha') (2\partial_+ \partial_- + \partial_i^2 + \sum_{i=1}^{D-2} -\partial_i^2 \partial_i^2)$

$$i \frac{\partial}{\partial \tau} = \alpha' (2\partial_+ \partial_-)$$

.... Not sure about this ... We prove this with...

$$\partial_{\tau} = 2\alpha' p^+ \partial_+$$

I think ... $p^+ = \frac{1}{i} \frac{\partial}{\partial x^-}$ and so...

$$\partial_{\tau} = 2\alpha' \frac{1}{i} \partial_- \partial_+ \Leftrightarrow i \partial_{\tau} = 2\alpha' \partial_- \partial_+ \quad \checkmark$$

Think this has to do w/ $[q^-, p^+]$ and
 x^- having to be dual to p^+ .

Looking at D.O.F.:

* There are no constraints and so all the D.O.F.

in 4 are physical states.

• Mass is given by: $\frac{1}{\alpha'} \left(-\alpha_0 + \sum_{i=1}^{D-1} \sum_{n=1}^{\infty} \alpha_n^{i\dagger} \alpha_n^i \right) \approx m^2$

• We now look at the mass spectrum when $\alpha_0 = 1$
 * The lowest mass state ϕ has

$$(m_{\text{mass}})^2 \sim -1/\alpha'$$

Why? because $\sum \alpha_n^{i\dagger} \alpha_n^i \geq 0$ so the minimum occurs (from ϕ) when $= 0$, Thus

$$m_{\text{min}}^2 \approx -1/\alpha', \quad \alpha_0 = 1$$

• Massless states: Classified by reps of the little group $SO(D-2)$ $p^\mu = (p, 0, 0, \dots, 0)$

• Massive states: reps of $SO(D-1)$
 $p^\mu = (p, 0, \dots, 0, 0)$

• Since $\psi = (\phi(x^\mu) + i \alpha_1^{\dagger i} A_i(x^\mu) + \dots) \psi_0$
 we have $[i=1, \dots, D-2]$ $D-2$ states A_i
 which can belong to the vector rep of $SO(D-2)$

• A_i : Massless vectors of $SO(D-2)$

$\alpha_n^i, i=1, \dots, D-2$ transforms as a rotation under $SO(D-2)$

* Level 2: $\psi^{(2)} \approx i \alpha_1^{i\dagger} \alpha_1^{j\dagger} B_{ij}(x^\mu) + i \alpha_2^{\dagger i} C_i(x^\mu)$

• Thus, if we count all the states of Level 2

$$|B_{ij}| + |C_i|, \quad i, j = 1, \dots, D-2$$

Since

$$[\alpha_n^i, \alpha_m^j] = n \delta^{ij} \delta_{n+m, 0}$$

$n=m=1$ and so

$$[\alpha_1^i, \alpha_1^j] = 0 \Rightarrow \alpha_1^i \alpha_1^j = \alpha_1^j \alpha_1^i$$

So B_{ij} is symmetric

$$B_{ij} = B_{ji}$$

of elements in an symm. matrix:

$$\begin{pmatrix} * & \dots & * \\ & \ddots & \\ & & * \\ n & & & n \end{pmatrix} \quad (n^2 - n)/2$$

Thus, the total # of independent elements in B_{ij} is $(n^2 - n)/2 + n$

$$|B_{ij}| = \frac{n^2 + n}{2}$$

Here $n = D - 2$, hence

$$\begin{aligned} [(D-2)^2 + (D-2)]/2 &= \{(D-2)(D-2) + 1\}/2 \\ &= (D-2)(D-1)/2 \end{aligned}$$

• Thus, we find: $|B_{ij}| = \frac{(D-2)(D-1)}{2}$

The total number of elements of level 2:

$$\begin{aligned} |B_{ij}| + |C_i| &= (D-2) + \frac{(D-2)(D-1)}{2} = \frac{(D-2)(2 + D-1)}{2} \\ &= (D-2)(D+1)/2 \end{aligned}$$

• $|B_{ij}| + |C_i| = \frac{(D-2)(D+1)}{2}$ as reps of $SO(D-2)$

* Now, t_{ij} : symmetric traceless rep of $SO(D-1)$, $i, j = 1, \dots, D-1$
 $t_{ii} = 0$ has the following number of independent components:

$$|t_{ij}| = \frac{n^2 + n}{2} - 1 = \frac{n^2 + n - 2}{2} = \frac{1}{2}(n-1)(n+2)$$

$n = D - 1$

$$|t_{ij}| = \frac{1}{2}(D-2)(D+1) = \frac{(D+1)(D-2)}{2}$$

* This is exactly the same number of D.O.F.: we therefore embed this into the massive states of $SO(D-1)$:

$$\begin{pmatrix} B_{ij} & i, j = 1, 2, \dots, D-2 \end{pmatrix}$$

$$t_{ij} = \begin{cases} c_i & i=1, \dots, D-2, i=D-1 \end{cases}$$

$$t_{ij} = \begin{pmatrix} B_{11} & \dots & B_{D-2, D-2} & C_1 \\ & B_{22} & & \vdots \\ & \vdots & \ddots & \vdots \\ B_{D-2, D-2} & & & C_{D-2} \\ C_1 & \dots & C_{D-2} & \psi \end{pmatrix}$$

$$\psi = t_{D-1, D-1}$$

• To determine ψ we note $\text{tr}(t_{ij}) = 0$

$$\text{tr}(t_{ij}) = \sum_{i=1}^{D-2} B_{ii} + \psi = 0$$

I'm assuming that either the SOCP-1) extension uses A_{ij} or there is a typo and $\psi = -\sum_i B_{ii}$.

→ West has $\psi = \sum_i B_{ii}$.

An aside from Polchinski:

* States of the theory: Let us expand in Dirac Notation what is meant by

$$\psi = (\phi_0 + \alpha^i A_i + \dots) \psi_0$$

→ A state $|0; k\rangle$, $k = (k^+, k^i)$ is defined as eigenstates of the creation operators $\alpha_m^\dagger$:

$$p^+ |0; k\rangle = k^+ |0; k\rangle, \quad p^i |0; k\rangle = k^i |0; k\rangle$$

$$\alpha_m^i |0; k\rangle = 0, \quad m > 0$$

• General states are built from

$$(\alpha_m^i)^\dagger |0; k\rangle$$

$$\text{Since } (\alpha_m^i)^\dagger = \alpha_{-m}^i:$$

$$|N; k\rangle = \left[\prod_{i=2}^{D-2} \prod_{n=1}^{\infty} \frac{(\alpha_n^i)^{N_i}}{(n N_i N_i!)^{1/2}} \right] |0; k\rangle$$

- $N = \text{occupation \#s } (i, n)$. Every occupation # corresponds to a different particle or spin state

- $\mathcal{H} = \text{Span}_N \{ |N; k\rangle \}$

The state $|0; 0\rangle$ is not the vacuum state but rather the state of a string w/ no momenta

$$H = \frac{p^\mu p_\mu}{2p^+} + \frac{2}{p^+ \alpha'} \left(\sum_{n=1}^{\infty} \alpha_{-n}^i \alpha_n^i + A \right)$$

→ This differs from last

$$2p^+ \alpha' \text{flues} = \text{florichinski}$$

General state at level n :

$$\alpha_{m_1}^{i_1} \dots \alpha_{m_p}^{i_p} \zeta_{i_1 \dots i_p} \zeta_0 \quad \text{s.t.} \quad \sum_i m_i = n$$

$n=0$: $|0; k\rangle$

$n=1$: $\alpha_{m_i}^{i_1} A_{i_1} \quad \alpha_{-1}^i |0; k\rangle$

$n=2$: $\alpha_2^i C_i + \alpha_1^{i_1} \alpha_1^{i_2} A_{i_1 i_2} \quad \alpha_{-2}^i |0; k\rangle, \alpha_{-1}^{i_1} \alpha_{-1}^{i_2} |0;$

$n=3$: $\alpha_3^i C_i + \alpha_1^i \alpha_1^{j_1} \alpha_1^{j_2} B_{ij_1 j_2} + \alpha_2^i \alpha_1^{j_1} K_{ij_1}$

• As we stated lets look at this

level:

$$C_i: D-2$$

$$K_{ij}: ((D-2)^2 + (D-2)) / 2 = \frac{(D-2)(D-1)}{2}$$

$B_{ij_1 j_2}$: Symmetric 3-tensor

• $\square\square\square: \frac{(D-2)(D-1)D}{3 \cdot 2}$

$$\dim(\square\square\square) = \frac{(D-2)(D-1)D}{6}$$

... Note that: $\dim(V) = N$

$$\dim(\text{Sym}^k(V)) = \binom{N+k-1}{k}$$

$$N = D-2, \quad k = 3$$

$N = D-2, K=3$

$$\binom{D}{3} = \frac{D!}{3!(D-3)!} = \frac{D(D-1)(D-2)}{6}$$

$$\begin{aligned} \therefore |C_i| + |K_{ijk}| + |B_{ij}| &= (D-2) + \frac{(D-2)(D-1)D}{6} + \frac{(D-2)(D-1)}{2} \\ &= \frac{D-2}{6} (D^2 + 2D + 3) = \frac{1}{6} (D^3 - D - 6) = \frac{1}{6} D(D^2 - 1) - 1 \\ &= \frac{1}{6} D(D+1)(D-1) - 1 \end{aligned}$$

Note $\dim(\text{Sym}^3(M^{D-1})) = \binom{D+1}{3} = \frac{(D+1)D(D-1)}{6}$

$$\therefore \underbrace{\frac{1}{6} D(D+1)(D-1)}_{\substack{\text{Symmetric} \\ 3\text{-tensor} \\ C_{ijk}}} \underbrace{- 1}_{\substack{\text{Not sure ...} \\ \text{trivial rep.}}}$$

• Again, if a state at level n is given by $\psi_{i_1} \dots \psi_{i_n}$, $\sum_i m_i = n$

$c(n) \equiv (\# \text{ of such states at level } n)$

Claim: $c(n)$ can be determined by the formula

$$\sum_n c(n) x^n = \prod_{n=1}^{\infty} \frac{1}{(1-x^n)^{D-2}}$$

RHS: $\frac{1}{(1+x)^{D-2}} \frac{1}{(1+x^2)^{D-2}} \dots$

$$\frac{1}{(1+x)^{D-2}} = \sum_n \frac{f^{(n)}(0)}{n!} x^n \quad \frac{d^3}{dx^3} () = D(D-1)(D-2)$$

$$\frac{d}{dx} () = (D-2) \frac{1}{(1-x)^{D-1}}$$

$$\frac{d^2}{dx^2} (1) = (D-2)(D-1) \frac{1}{(1-x)^2}$$

$$1 + \frac{(D-2)(D-1)}{2} x^2 + (D-2)x^2 + (D-2)x^3 + \frac{D(D-1)(D-2)}{6} x^3 + (D-2)x^2 + (D-2)^2 x^3$$

$$\left(1 + \frac{D(D-1)(D-2)}{6} x^3 + \frac{(D-2)(D-1)}{2} x^2 + x(D-2) \right) (1 + (D-2)x^2) (1 + (D-2)x^3)$$

$$x^3: (D-2) + \frac{D(D-1)(D-2)}{6} + (D-2)^2$$

$\underbrace{\hspace{10em}}$
 have
 $(D-2)(D-1)/2$ instead

* Assuming $(D-2)^2$:

$$(D-2) \left((D-1) + \frac{D(D-1)}{6} \right) = \left[(D-2)(D-1)/6 \right] (6 + D-1)$$

$$= \frac{(D+5)(D-2)(D-1)}{6}$$

Interesting! I got this by assuming there were two sym. tensors:

$$x_2 x_1 \text{ and } x_1 x_2$$

$$\text{So that } (D-2)(D-1)/2 \rightarrow (D-2)(D-1)$$

--

$$|C_i| + |K_{ij}K| + 2|B_{ij}|$$

$$(D-2) + \frac{D(D-1)(D-2)}{6} + (D-2)(D-1)$$

$$= \frac{(D-2)D}{6} (6 + D-1) = \frac{D(D+5)(D-2)}{6}$$

err... nevermind. My answer differs from D and D-1...

See my "extra" notes for a full analysis!

* Proof: Harmonic Oscillator: $[a, a^\dagger] = 1$

Has one eigenstate of each level

$$|n\rangle = (a^\dagger)^n / \sqrt{n!} |0\rangle$$

$$a|0\rangle = 0$$

• The number of energy states of level n :

$$\sum_{n=0}^{\infty} \tilde{Z}(n) x^n = \frac{1}{(1-x)} = 1 + x + x^2 + x^3 + \dots$$

$$\tilde{Z}(n) = 2 \quad \forall n$$

* This equation can be written in terms of the trace over the Hilbert-Space of states:

→ For a state $|n\rangle$ the number of states can be determined by the number operator

$$\hat{N} \equiv a^\dagger a$$

Then

$$x^{\hat{N}} |n\rangle = x^n |n\rangle$$

Since $\{|n\rangle\}$ span a complete basis of the Hilbert Space we can trace over them

$$\text{Tr}(x^{a^\dagger a}) = \sum_n \langle n | x^{a^\dagger a} | n \rangle = \sum_n x^n \quad \checkmark$$

$$\boxed{\text{Tr}(x^{a^\dagger a}) = \sum_n x^n = \frac{1}{1-x}}$$

• We can also expand this as (using:

$$\log(\det(\mathcal{O})) = \text{tr}(\ln(\mathcal{O}))$$

$$x^{a^\dagger a} = e^{\ln(x^{a^\dagger a})} = \exp[a^\dagger a \ln(x)]$$

$$\text{tr}(x^{a^\dagger a}) = \text{tr}(\exp[a^\dagger a \ln(x)])$$

• For the 1D-string $\alpha_n \equiv \alpha_n^I$, $n=1, \dots, 2$

$$\psi = (\phi + i\alpha_1 A_{(1)} + \alpha_1 \alpha_2 B_{(2)} + \alpha_2 A_2 + \dots$$

$$n=0: 1 \quad + \alpha_1 \alpha_2 + \alpha_1 \alpha_1 \alpha_1 + \alpha_3$$

$$n=1: 2$$

$$n=2: 2 \quad n=4: 4$$

$$n=3: 3$$

$$\left(\frac{1}{1-x}\right) \left(\frac{1}{1-x^2}\right) \sim (1+x+x^2+\dots)(1+x^2)(1+x^3)$$

$$\sim (1+x+x^2+x^3+x^2+x^3+x^3)$$

$$\sim (1 + x + 2x^2 + 3x^3 + \dots)$$

In this case there are n states for the level n . As we can see, this is given

$$\text{by } \sum_n \hat{C}(n) x^n = \prod_{n=1}^{\infty} \frac{1}{(1-x^n)}, \quad \hat{C}(n) = n$$

The states of level n are of the form

$$\sum_{p_1, \dots, p_p} \alpha_{n_1}^\dagger \dots \alpha_{n_p}^\dagger |n_1, \dots, n_p\rangle \delta(n_1 + \dots + n_p - n) \psi_0$$

• Here $\hat{N} \equiv \sum_n \alpha_{-n} \alpha_n$

$$\sum_n C(n) x^n = \text{Tr}(x^{\hat{N}})$$

$$\begin{aligned} D &= 1 : \sum_n C_n x^n = \text{Tr}(x^{\sum_n \alpha_{-n} \alpha_n}) = \text{Tr}(\exp(\sum_n \alpha_{-n} \alpha_n \ln(x))) \\ &= \text{Tr} \left\{ \prod_{n=1}^{\infty} \exp(\alpha_{-n} \alpha_n \ln(x)) \right\} \\ &= \prod_{n=1}^{\infty} \text{Tr}^{(n)}(\exp(\alpha_{-n} \alpha_n \ln(x))) \end{aligned}$$

$$\text{Since } \exp(\alpha_{-n} \alpha_n) = \mathbb{I} + \underbrace{\alpha_{-n} \alpha_n}_{\mathcal{H}^{(1)}} + \frac{1}{2} \underbrace{(\alpha_{-n} \alpha_n)^2}_{\mathcal{H}^{(2)}} + \dots$$

we can look at tensor states

$$\mathcal{H}^{(2)} \cong \mathcal{H}^{(1)} \otimes \mathcal{H}^{(1)}$$

$$\mathcal{H}^{(n)} \cong \mathcal{H}^{\otimes n}$$

and so the trace is done over this tensor state. This is the meaning of $\text{tr}^{(n)}$

$$\alpha_n \equiv \sqrt{n} a_n, \quad \alpha_{-n} \equiv \sqrt{n} a_{1-n}$$

for which

$$[a_n, a_m^\dagger] = \delta_{n+m, 0} \quad n > 1.$$

* Now, $\sum_n C_n x^n = \prod_{n=1}^{\infty} (\text{Tr}^{(n)} \{ \exp(\alpha_{-n} \alpha_n \ln(x)) \})$

$$\prod_{n=1}^{\infty} = \prod_{n=1}^{\infty} \left(\text{Tr}^{(n)} \{ \exp(\alpha_{-n} \alpha_n \ln(x)) \} \right)$$

$$= \prod_{n=1}^{\infty} \text{Tr}^{(n)} \{ \exp(n a_{-n} a_n \ln(x)) \} = \prod_{n=1}^{\infty} \text{Tr}^{(n)} \{ \exp(a_{-n} a_n \ln(x^n)) \}$$

$$= \prod_{n=1}^{\infty} \frac{1}{1-x^n}$$

• This is how to formally obtain the claim.
Now, we have $D=2$.

$$\underline{D \rightarrow D-2} : \quad \hat{N} \equiv \sum_{n=1}^{\infty} \sum_{i=1}^{D-1} \alpha_{-n}^i \alpha_n^i$$

$$\begin{aligned} \sum_n C(n) x^n &= \text{Tr}(x^{\hat{N}}) = \text{Tr}(\exp\{\hat{N} \ln(x)\}) \\ &= \text{Tr}\left(\prod_{n=1}^{\infty} \prod_{i=1}^{D-2} \exp\{\alpha_{-n}^i \alpha_n^i \ln(x)\}\right) \\ &= \prod_{n=1}^{\infty} \prod_{i=1}^{D-2} \text{Tr}^{(n,i)}(\exp\{a_{-n}^i a_n^i \ln(x^n)\}) \\ &= \prod_{n=1}^{\infty} \underbrace{\text{Tr}^{(n,1)}} \text{Tr}^{(n,2)} \dots \text{Tr}^{(n,D-2)} \end{aligned}$$

Each of these yield the same result:

$$\text{Tr}^{(n,i)} = \frac{1}{1-x^n} \quad \forall i=1, \dots, D-2$$

Thus:

$$\sum_n C(n) x^n = \prod_{n=1}^{\infty} \frac{1}{(1-x^n)^{D-2}}$$

* Light-cone states must have $\| |n\rangle \| > 0$ since they are created by $\alpha_n^{i\dagger}$ and obey $[\alpha_n^i, \alpha_m^{j\dagger}] = n \delta_n^m \delta_i^j$
 $\rightarrow$ Thus we get no ghosts

Lorentz Symmetry & $D=26$:

* In the previous section we have $\alpha_0 = 1$
 Since we require A_i to transform as a massless vector of $SO(D-2)$

$$M^2 = 2(\hat{N} - a_0)/\alpha'^2$$

• The first excited state $A_i = \alpha_{-1}^i |0, p\rangle$ must be massless.

Now,

$$\hat{N} |A_i\rangle = |A_i\rangle$$

i.e. there is only 1 state.

$$\therefore M^2 |A_i\rangle = 0 \Rightarrow \hat{N} |A_i\rangle = \alpha_0 |A_i\rangle = |A_i\rangle$$

and so

$$\boxed{\alpha_0 = 1}$$

* Recall: $T^{\mu\nu} = \int_a^\pi M_0^{\mu\nu} d\sigma = \int_a^\pi (x^\mu p^\nu - x^\nu p^\mu) d\sigma$
For the open string $a=0$

i.e.:

$$J^{ij} = \int_0^\pi d\sigma (x^i p^j - x^j p^i)$$

* An open string w/ Neumann B.C.'s is given by:

$$X^\mu(\tau, \sigma) = x^\mu + \sqrt{2\alpha'} \alpha_0^\mu \tau + i \sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} (\alpha_n^\mu e^{-in(\tau+\sigma)} + \alpha_n^\mu e^{-in(\tau-\sigma)})$$

$$J^{ij} = \frac{1}{2\alpha'} \sqrt{\frac{\alpha'}{2}} \int_{-\pi}^\pi d\sigma (x^\mu p^\mu - x^\nu p^\nu)$$

$$\boxed{J^{ij} = \frac{1}{\sqrt{2}\alpha'} (x_0^i \alpha_0^j - x_0^j \alpha_0^i) + \sum_{n=1}^{\infty} \frac{(\alpha_n^i \alpha_n^j - \alpha_n^j \alpha_n^i)}{in}}$$

$$J^{i+} = \int_0^\pi d\sigma (x^i p^+ - x^+ p^i) =$$

$$= \int_0^\pi d\sigma (x_0^i + \sum_{n \neq 0} \alpha_n^i e^{in\sigma} |p^+/\pi| - (x_0^+ + c\tau) p_0^i)$$

$$p^i = \frac{1}{\pi} \sum_{n=0}^{\infty} p_n^i e^{in\sigma} \quad \text{only } p_0^i \text{ survives}$$

$$J^{i+} = x_0^i p^+ - (x_0^+ + c\tau) p_0^i$$

$$C = 2\alpha' p^+$$

$$J^{i+} = x_0^i \tau p^+ - (x_0^+ + 2\alpha' p^+ \tau) p_0^i$$

$$x_0^i(\tau) = x_0^i(0) + 2\alpha' p^i \tau$$

$$J^{i+} = (x_0^i(0) + 2\alpha' p_0^i \tau) p^+ - x_0^+ p_0^i - 2\alpha' p^+ p_0^i$$

$$= \underbrace{x_0^i(0) p^+}_{\uparrow} - x_0^+ p^+ \quad \text{Not sure why this isn't part of soln.}$$

$$J^{+-} = \int_0^\pi d\sigma (x^+ p^- - x^- p^+)$$

• apparently $x_0^+ = 0$? ... $x^+ = 2\alpha' \tau p^+$

$$J^{+-} = \int_0^\pi d\sigma (2\alpha' \tau p^+ p^- - x^- (\frac{p^+}{\pi}))$$

$$= 2\alpha' \tau p^+ \int_0^\pi d\sigma p^- - p^+ \frac{1}{\pi} \int_0^\pi d\sigma x^- (\tau, \sigma)$$

where

$$H = -2\alpha' p^+ \int_0^\pi d\sigma p^-(\sigma)$$

$$J^{+-} = -H\tau - p^+ q^-(\tau)$$

$$q^- = q^-(0) - (H/p^+) \tau$$

$$\therefore J^{+-} = -p^+ q^-(0)$$

* Collecting terms:

$$J^{ij} = \frac{1}{2\alpha'} (x_0^i x_0^j - x_0^j x_0^i) + \sum_{n=1}^{\infty} \frac{(\alpha_{-n}^i \alpha_n^j - \alpha_{-n}^j \alpha_n^i)}{in}$$

$$J^{i+} = p^+ x_0^i(0)$$

$$J^{+-} = -p^+ q^-(0)$$

Angular
Momentum
from
Lor. Square

* Need to calculate J^{i-} :

$$J^{i-} = \int_0^\pi d\sigma (x^i p^- - x^- p^i)$$

Let us introduce oscillators α_n , defined

by:

$$\begin{aligned} p^-(\tau, \sigma) &\equiv \frac{1}{\sqrt{2\alpha'}} (\dot{X}^- + (\dot{X}^-)') = \pi \sqrt{2\alpha'} P^- + (\dot{X}^-)' / \sqrt{2\alpha'} \\ &= \sum_n \alpha_n^- e^{-in\sigma} \end{aligned}$$

To relate these to α_n^i we note:

$$0 = \eta^{\mu\nu} p_\mu = 2p^+ p^- + \sum_{i=1}^{D-2} p^i p_i$$

$$p^+ = \frac{1}{\sqrt{2\alpha'}} (\dot{X}^+ + (\dot{X}^+)') = \pi \sqrt{2\alpha'} P^+ = \sqrt{2\alpha'} p^+$$

$$p^- = - \frac{1}{2p^+ \sqrt{2\alpha'}} \sum_{i=1}^{D-2} p_i^2 = \sum_n \alpha_n^-(\tau) e^{-in\sigma}$$

Recall

$$p^i(\tau, \sigma) = \sum_n \alpha_n^i e^{-in\sigma}$$

$$\left(-\frac{1}{2p^+ \sqrt{2\alpha'}} \right) \sum_{l,m} \alpha_l^i \alpha_m^i e^{-i(k+l+m)\sigma} = \sum_n \alpha_n^- e^{-in\sigma}$$

$$\alpha_n^- = \left(-\frac{1}{2p^+ \sqrt{2\alpha'}} \right) \sum_{l=1}^{D-2} \sum_m \alpha_{n-l}^i \alpha_m^i$$

$$\therefore \dot{X}^- \approx \frac{1}{\sqrt{2\alpha'}} (X_0^i \alpha_0^- - X_0^- \alpha_0^i) + \sum_{n=1}^{\infty} \frac{(\alpha_{-n}^i \alpha_n^- - \alpha_n^- \alpha_{-n}^i)}{in}$$

* Now compute the variation of $X^i(\tau, \sigma)$ under a Lorentz transf. Classically:

$$\delta X^i(\tau, \sigma) = \left\{ X^i(\tau, \sigma), \frac{\omega_{\mu\nu}}{2} J^{\mu\nu} \right\}$$

$$e^{\frac{\omega_{\mu\nu}}{2} J^{\mu\nu}} X^i = X^i + \delta X^i$$

• Under ω^{ij} we have

$$\delta X^i = -\omega^{ij} X_j = -\omega^{ij} x_j$$

